Master of Science in Software Engineering
May 2026

Human–AI Collaboration for Decision-Making in Agile Sprint Planning A Practitioner-Informed Conceptual Model

Ungarala Sai Krishna Yashwanth Mihir Singh Thakur

Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden

This thesis is submitted to the Faculty of Computing at Blekinge Institute of Technology in partial fulfillment of the requirements for the degree of Master of Science in Software Engineering. The thesis is equivalent to 20 weeks of full-time studies.

The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

Contact Information:
Author(s):
Ungarala Sai Krishna Yashwanth
E-mail: saun24@student.bth.se

Mihir Singh Thakur
E-mail: mith24@student.bth.se

University advisor:
Henry Edison, PhD
Department of Software Engineering

Faculty of Computing
Blekinge Institute of Technology
SE–371 79 Karlskrona, Sweden

Internet : www.bth.se
Phone : +46 455 38 50 00
Fax : +46 455 38 50 57

Abstract

Background. Sprint planning is a critical decision-making process in Agile software development. AI-powered tools increasingly offer recommendations during planning, yet how practitioners actually respond to these recommendations remains poorly understood. Two failure patterns are relevant: anchoring bias, where teams adjust insufficiently from an AI-suggested figure, and algorithm aversion, where practitioners reject valid recommendations without evaluation. Neither pattern has been examined empirically at the practitioner-interaction level in sprint planning contexts.
Objectives. This thesis develops a practitioner-informed conceptual model for Human-AI collaboration in Agile sprint planning, designed to reduce cognitive bias risk in decision-making while preserving team autonomy and human oversight.
Methods. We conducted semi-structured interviews with nine Agile practitioners across developer, Scrum Master, tech lead, and engineering manager roles, embedding two vignette scenarios to surface anchoring and algorithm aversion in participant reasoning. Thematic analysis following Braun and Clarke's six-phase framework was applied to all transcripts. The coding framework was structured around the four meta-dimensions of the HACO taxonomy.
Results. Anchoring bias was the most consistently identified pattern, with five of nine participants independently describing number-first contamination as a risk they actively guard against. Algorithm aversion appeared in a more nuanced form than prior literature suggests: most experienced practitioners showed calibrated scepticism rather than outright rejection. The strongest emergent finding was a shared interaction model in which the AI tool informs the facilitator privately before the session, and the facilitator then decides what to bring to the team and how to frame it, preserving independent team estimation. Trust was found to be conditional primarily on AI transparency of reasoning.
Conclusions. The conceptual model specifies three components: role definitions separating AI analytical support, Human-AI dialogue, and human-only zones; five interaction principles derived from practitioner experience; and a risk identification checklist. The most strongly supported principle is number withholding: keeping AI-suggested figures from the team until after independent estimation. The model offers a practitioner-grounded starting point for structuring Human-AI collaboration in sprint planning contexts.

Keywords: Human-AI collaboration, Agile sprint planning, cognitive bias, anchoring bias, algorithm aversion, calibrated trust, qualitative research, vignette methodology.

Acknowledgements

We thank our supervisor, Henry Edison, for his oversight of this project throughout the course.

We also thank the nine practitioners who gave their time to participate in the interviews. Their candid accounts of sprint planning practice, AI tool adoption, and decision-making challenges are the foundation of this work.

Finally, we thank our families and peers for their support during a demanding semester.

Abbreviations

AI	Artificial Intelligence
BTH	Blekinge Institute of Technology
HACO	Human-AI Collaboration Framework
LLM	Large Language Model
RQ	Research Question
TA	Thematic Analysis

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background

Sprint planning sits at the heart of every Agile development cycle. It is the moment where a team translates a backlog of work into a concrete commitment, deciding what will be done, by whom, and within what timeframe. Getting this right matters: research shows that poor sprint planning leads directly to incomplete sprints, wasted capacity, and declining team confidence [11, 13].

What makes sprint planning difficult is not a lack of information but too much of it, under time pressure. Teams must weigh developer availability, story complexity, technical debt, stakeholder priorities, and past performance within a timeboxed session [13, 19]. Under these conditions, shortcuts in reasoning are not a sign of incompetence. They are a natural response to cognitive overload: when deliberate analytical thinking becomes too costly under time pressure, the mind defaults to faster, heuristic-driven reasoning [17, pp. 20–24]. These shortcuts show up as predictable patterns that researchers call cognitive biases [6, 21].

AI-powered project management tools now offer sprint recommendations based on historical data [3, 5]. The assumption behind most of these tools is straightforward: if the AI tool has processed more data than any individual team member, its recommendations should improve planning quality. But this assumption sidesteps a critical question: how do practitioners actually respond to these recommendations in practice?

Research on Human-AI decision-making suggests the answer is often more complicated than expected. Two failure patterns appear repeatedly. The first is *anchoring bias*: when an AI tool produces a figure, teams tend to treat it as a reference point and adjust insufficiently from it, even when their own experience should lead them to a different conclusion [21][17, pp. 119–128]. The second is *algorithm aversion*: practitioners who distrust AI tools, or feel their professional judgement is being undermined, reject valid recommendations without evaluation [10]. Both patterns degrade decision quality in opposite directions and are well documented in the literature [10, 21].

Recent Human-AI collaboration research increasingly emphasises trust calibration, role clarity, and interaction design as the key factors determining whether AI assistance actually helps [2, 16, 22]. Prior work covers AI-generated sprint plans, story point estimation, and LLM-based planning support [8, 19, 22], but to the best of our knowledge no prior study simultaneously examines cognitive bias mechanisms at the practitioner-interaction level, trust calibration dynamics, and derives a practitioner-informed collaboration model specifically within Agile sprint planning contexts. That gap is what this thesis addresses.

Our central organising concept is *calibrated trust*. Lee and See [20] define appropriate reliance as trust that is proportional to a system's actual capabilities, neither over-relying nor under-relying. A team's reliance on AI recommendations should match how reliable those recommendations actually are in a given situation.

1.2 Aim and Objectives

The aim of this thesis is to develop a practitioner-informed conceptual model for Human-AI collaboration in Agile sprint planning that reduces cognitive bias risk while preserving team autonomy and human oversight.

To achieve this aim, the thesis pursues four objectives:

1. Examine how anchoring bias and algorithm aversion affect decision quality when AI recommendations are introduced into Agile sprint planning.

2. Examine how these biases manifest in practitioner reasoning during AI-assisted planning scenarios.

3. Analyse how Agile practitioners calibrate trust in AI tool recommendations and what conditions lead to overtrust or distrust.

4. Construct a conceptual model specifying role definitions, interaction principles, and risk conditions for Human-AI collaboration in sprint planning.

1.3 Problem Statement

This thesis investigates how cognitive biases affect practitioner decision-making during AI-assisted sprint planning, how trust in AI tools is calibrated in that context, and what principles should govern Human-AI collaboration to reduce bias risk while preserving human oversight.

1.4 Research Questions

Four research questions guide the study:

- **RQ1a (Diagnostic):** How do anchoring bias and algorithm aversion affect decision quality in Agile sprint planning when AI recommendations are introduced?

- **RQ1b (Diagnostic):** How do these biases manifest in practitioner reasoning during AI-assisted sprint planning?

- **RQ2 (Relational):** How do Agile practitioners perceive and calibrate their trust in AI tools when delegating planning tasks, and what conditions lead to overtrust or distrust?

- **RQ3 (Constructive):** What principles should govern Human-AI collaboration in Agile sprint planning to reduce cognitive bias risk while maintaining human oversight and team autonomy?

RQ1a and RQ1b examine what already goes wrong in sprint planning before any model is proposed. RQ2 examines the Human-AI relationship dynamic that either enables or undermines effective collaboration. RQ3 turns outward: given what we learn from RQ1 and RQ2, what should a collaboration model specify?

1.5 Thesis Outline

Chapter 2 reviews existing work on Human-AI collaboration, cognitive bias in software engineering, and AI in Agile development. Chapter 3 describes the research design, participant recruitment, data collection, and analysis approach. Chapter 4 presents the findings and the resulting conceptual model. Chapter 5 interprets the findings in relation to existing theory and addresses limitations. Chapter 6 answers the research questions directly, states the contributions, and outlines directions for future work.

Chapter 2

Related Work

2.1 Human-AI Collaboration

Recent Human-AI collaboration research increasingly emphasises trust calibration and role clarity as the factors that most reliably distinguish effective teaming from ineffective teaming. A systematic review by Aman et al. [2] found that these two factors appear consistently across different task domains and team configurations. Emmanouilidis et al. [14] proposed co-creation design patterns for Human-AI collaboration in decision-making contexts, arguing that the design of interaction protocols, not the AI tool's predictive accuracy, is what determines whether collaboration produces better outcomes than either humans or AI tools working alone.

Within software engineering, Klieger et al. [18] found that when humans and AI tools collaborate on software tasks, the quality of outcomes depends heavily on how well roles and responsibilities are defined and how much transparency the AI tool provides about its reasoning. A workshop study on ChatGPT in software engineering contexts [15] found that practitioners were generally willing to engage with AI suggestions but needed explicit mechanisms to evaluate them before committing.

2.2 AI in Agile Sprint Planning

Muneer [22] compared different AI architectures for sprint planning support and found that LLM-based approaches produced more useful outputs than traditional optimisation models, but that human oversight remained essential because AI tools could not account for team context. Hamza et al. [16] found that AI integration in Agile project management improved certain metrics but introduced new challenges around practitioner trust and resistance. Neither study produced a collaboration model or investigated the cognitive dynamics of how practitioners engage with AI recommendations in the moment of planning.

Prior work also addresses story point estimation [8] and context-aware sprint plan generation [19]. Barua et al. [5] and Angara et al. [3] document the growing availability of AI-powered project management tools. None of this work, however, examines the cognitive mechanisms through which practitioners accept or reject AI recommendations during planning.

2.3 Cognitive Bias in Software Engineering

Borowa et al. [6] documented fixation patterns in Agile teams that share structural similarities with anchoring behaviour. A systematic mapping of cognitive biases in software engineering found anchoring and confirmation bias to be among the most frequently observed [21]. Drury-Grogan and O'Dwyer [11] showed that time pressure during sprint planning significantly increases the likelihood that teams revert to heuristic reasoning rather than deliberate analysis [17, pp. 20–24].

Anchoring bias occurs when an initial piece of information serves as a reference point from which a person adjusts insufficiently [17, pp. 119–128]. In sprint planning, an AI recommendation that arrives before team discussion constitutes such an anchor [19]. *Algorithm aversion* is the tendency to avoid algorithmic recommendations disproportionately after observing them err, even when the algorithm remains more accurate than human judgement [10].

Complacency, documented by Parasuraman and Manzey [24], represents a related risk: over-reliance on automated output leads to reduced vigilance and skill degradation over time. Bao et al. [4] documented similar dynamics in Human-AI collaboration settings.

2.4 Frameworks for Human-AI Collaboration

The HACO framework proposed by Dubey et al. [12] provides a taxonomy for Human-AI teaming, organised across four meta-dimensions: task characteristics, covering goals, task allocation, and AI roles; learning paradigm, capturing how both human and AI learn from the interaction; teaming characteristics, covering team relationships, observability, predictability, shared awareness, and adaptability; and trust, covering calibrated trust and interpretability.

Lee and See [20] established the foundational theory of calibrated trust in automation, defining appropriate reliance as the target state for Human-AI tools. While HACO identifies trust as a relevant dimension, it does not explain how trust develops or what appropriate reliance looks like in practice. Lee and See supply that theoretical foundation.

Nisbett and Wilson [23] demonstrated that people consistently cannot report accurately on the cognitive processes driving their behaviour. Aguinis and Bradley [1] provide methodological justification for the vignette approach as a means of eliciting more valid data on bias than direct self-report.

2.5 Summary and Research Gap

Table 2.1 summarises the reviewed literature along the dimensions most relevant to this thesis.

Based on the reviewed literature, no study was found that combines empirical grounding in Agile sprint planning, examination of cognitive bias mechanisms at the practitioner-interaction level, trust calibration analysis, and a practitioner-informed collaboration model. This thesis addresses that combination directly.

Table 2.1: Summary of related work and research gap

Study	Agile context	Cognitive bias	Trust calibration	Practitioner model
Aman et al. [2]	No	No	Yes	No
Emmanouilidis et al. [14]	No	No	Partial	No
Klieger et al. [18]	No	No	Partial	No
Muneer [22]	Yes	No	No	No
Hamza et al. [16]	Yes	No	Partial	No
Mohanani et al. [21]	Partial	Yes	No	No
Dubey et al. [12]	No	No	Yes	No
This thesis	**Yes**	**Yes**	**Yes**	**Yes**

Chapter 3

Method

3.1 Research Design

We chose a qualitative empirical design because the phenomena we investigate, cognitive bias and trust calibration, require context-sensitive data to understand how practitioners reason about AI recommendations in sprint planning [9, pp. 47–48]. A survey asking practitioners to rate their trust in AI on a scale of one to five tells us very little about what actually drives that trust, which situations trigger anchoring, or how reasoning shifts when an AI recommendation enters the room. To examine these dynamics, it is necessary to hear practitioners reason through concrete scenarios [11, 15].

We want to be transparent about our positionality as researchers. Both of us are MSc students in software engineering. We have studied AI tools and Agile methods academically, but neither of us has worked professionally as a Scrum Master, Product Owner, or Agile delivery lead, and neither has used an AI-assisted sprint planning tool in a real production environment. We approached this study as informed outsiders. This positionality has one advantage: we are less likely to project our own assumptions onto what participants tell us. The risk is that we may misread practitioner language or miss context that an insider would catch. We addressed this through member-checking in the validation walkthroughs and through a systematic coding process with explicit operational definitions applied consistently across all transcripts, helping to surface and resolve interpretive ambiguities before they became embedded in the analysis [7][9, pp. 47–48].

Semi-structured interviews were selected as the primary instrument. We considered alternative approaches including direct observation of sprint planning sessions, quasi-experimental designs with teams, and artefact-based case studies, but ruled these out due to access constraints and the exploratory nature of the research questions [9, pp. 86–87]. The vignette approach provides a methodologically sound alternative to direct self-report for studying cognitive bias [1, 23], and the verbal reasoning it elicits gives direct access to the cognitive processes under investigation. Nisbett and Wilson [23] demonstrated that people consistently cannot accurately report the cognitive processes driving their own behaviour. Asking participants directly whether they experience anchoring bias would therefore produce unreliable data. Placing participants in a concrete scenario and observing their verbal reasoning reveals patterns consistent with documented bias mechanisms without relying on self-report. This is the specific methodological justification for embedding vignettes within the semi-structured interview format, following best practice recommendations for vignette methodology in organisational research [1].

3.2 Participant Recruitment

We targeted 10 to 15 participants, with an initial recruitment goal of 20 to account for scheduling failures and drop-outs. A sample of 10 to 15 is consistent with established guidance for qualitative interview studies [9, pp. 146–149], and we continued recruitment until we reached thematic saturation [7]. No substantially new codes emerged after the seventh interview; subsequent interviews primarily reinforced and refined existing themes rather than introducing new categories. Table 3.1 shows the code emergence pattern across all nine interviews, demonstrating that all twelve codes were established by interview P7 and that P8 and P9 introduced no new codes.

Table 3.1: Code emergence across interviews supporting saturation claim (X = code present, – = code absent)

Code	P1	P2	P3	P4	P5	P6	P7	P8	P9
A1 Number contamination	–	–	X	X	–	X	X	–	–
A2 Reference adjustment	X	X	–	–	–	X	–	X	–
A3 Number withholding	–	–	X	X	–	–	X	X	–
B1 Dev authority	–	X	–	–	X	X	–	–	–
B2 Examine signals	–	–	–	–	–	–	–	X	–
B3 Pattern scepticism	–	–	–	–	–	X	X	X	–
B4 Contextual dismissal	X	X	–	X	X	X	X	X	X
C1 Transparency trust	–	–	–	–	–	X	X	–	–
C2 Track record	–	–	–	–	–	X	X	X	X
C3 Human final	X	X	–	–	X	–	X	–	X
C4 Facilitator mediator	–	–	–	–	–	X	X	–	–
C5 Contextual applicab.	–	–	X	–	–	–	–	X	–
New codes introduced	3	2	2	2	1	4	3	0	0

This pattern was taken as an indicator that thematic saturation had been reached.

An *Agile practitioner*, for the purposes of this study, is someone who currently participates in sprint planning as part of their job, with at least one year of sprint planning experience. This includes developers, Scrum Masters, Product Owners, and engineering managers. We did not require prior experience with AI planning tools, because a practitioner who has never used such a tool still offers valid perspective on the conditions under which adoption is resisted.

Recruitment proceeded through our professional and personal networks across the software industry. Before confirming a participant, we asked two screening questions: do they currently work in an Agile team, and do they participate in sprint planning. Both had to be answered affirmatively.

Table 3.2 provides an overview of the nine study participants. All identities and organisations have been anonymised in compliance with informed consent agreements.

Table 3.2: Participant overview

ID	Role	Experience	AI tool experience	Duration
P1	Engineering Manager	21 years	Yes	56 min
P2	Team Lead	5+ years	No	27 min
P3	Scrum Master	6+ years	Limited	58 min
P4	Scrum Master	5+ years	Yes (LinearB)	45 min
P5	Tech Lead	6+ years	Partial	38 min
P6	Scrum Master	8+ years	Yes	28 min
P7	Scrum Master	6+ years	Yes	73 min
P8	Scrum Master	6+ years	Yes	31 min
P9	Team Lead	13 years	Partial	39 min

3.3 Data Collection

All nine interviews were conducted online via video call between 18 April and 3 May 2026. Each interview ran for approximately 27 to 73 minutes depending on participant experience and depth of response.

The interview guide had three parts. Part 1 asked about the participant's current sprint planning practice: how their team runs planning sessions, what typically goes well, and where the difficulties usually are. This part was deliberately open to let participants describe their own experience before any scenarios were introduced.

Part 2 presented two vignette scenarios grounded in documented bias patterns from the literature [1, 10, 21]. Each scenario was designed to operationalise a specific bias mechanism identified in prior research, creating conditions under which participants would encounter that mechanism in their reasoning without being told what we were looking for.

Vignette A operationalised anchoring bias [17, 21]. Kahneman defines anchoring as the tendency to adjust insufficiently from an initial reference point [17, pp. 119–128]. Mohanani et al. [21] identify anchoring as one of the most frequently observed cognitive biases in software engineering, including in estimation contexts. The scenario presented an AI tool recommending 42 story points when the team's five-sprint average was 34, with a plausible but thin justification: three similar stories completed faster recently. The specific figures were not taken from a single paper but were calibrated following Aguinis and Bradley's guidance on ecological validity in vignette construction [1]: the scenario must be realistic and grounded in conditions practitioners actually encounter, and the target variable must be set at a level that requires genuine deliberation rather than being immediately obvious. A recommendation 24 percent above the team's established baseline, justified by a small number of recent fast completions, meets both criteria. It is plausible enough that a practitioner could reasonably accept it, and divergent enough that anchoring is detectable if present. If anchoring was present, participants would begin reasoning from 42 rather than forming an independent judgement.

Vignette B operationalised algorithm aversion [10]. Dietvorst et al. define algorithm aversion as the tendency to avoid algorithmic recommendations disproportionately after observing them err, even when the algorithm remains more accurate than human judgement. The scenario presented an AI tool flagging a story as likely to take 60% longer than estimated based on historical patterns, with the assigned developer disagreeing based on codebase familiarity. The 60 percent figure was chosen following the same ecological validity principle [1]: large enough to constitute a meaningful warning that a practitioner should engage with, but not so extreme as to make dismissal the only sensible response. The developer disagreement was included because Dietvorst et al. [10] specifically identify situations where human expertise conflicts with algorithmic output as the condition under which aversion is most likely to emerge. If algorithm aversion was present, participants would side with the developer without engaging with the AI tool's evidence base. It should be noted that Vignette B does not replicate the classic experimental precondition of prior observed algorithm error identified by Dietvorst et al. [10]. The scenario instead presents a conflict between an algorithmic signal and human expertise, designed to elicit responses relevant to algorithm aversion risk, particularly unjustified dismissal of algorithmic evidence. The study therefore does not claim to measure algorithm aversion in the strict experimental sense, but rather to examine the related pattern of dismissal without engagement with evidence.

Part 3 focused directly on trust: past experiences with AI tools, what drives acceptance or rejection of recommendations, and what information practitioners would need before delegating a planning decision to an AI tool [4, 20].

Informed consent was obtained verbally from each participant at the start of the interview. Each participant was informed of the study purpose, their right to withdraw at any time without consequence, and the anonymisation commitments outlined in the consent information sheet shared with each participant prior to the interview. All interviews were recorded with participant consent and transcribed within 48 hours of each session.

3.4 Data Analysis

We used thematic analysis to analyse the transcripts, following the six-phase framework established by Braun and Clarke [7]: familiarisation with the data; generation of initial codes; searching for themes; reviewing and refining themes; defining and naming themes; and producing the final analysis report.

The transcripts were coded by the primary researcher through open reading of all nine transcripts. The resulting codebook was reviewed with the second author, who applied the code definitions to five transcript excerpts independently; boundary cases were discussed and definitions refined before the codebook was applied to all remaining transcripts [9, pp. 251–252]. Where code definitions were ambiguous, particularly the boundary between B1 (developer authority assertion) and B4 (legitimate contextual dismissal), operational definitions were refined to require a concrete identifiable basis rather than assertion alone before the distinction could be applied.

The analysis combined open inductive coding in Phase A with deductive mapping in Phase B [7][9, pp. 251–252]. In Phase A, we applied open coding with no predetermined categories. In Phase B, we mapped the resulting themes onto the four HACO dimensions [12]. In Phase C, we constructed the conceptual model from the HACO-mapped themes.

Operational definitions used throughout coding: *Anchoring* was coded when a participant used the AI tool's figure as their first reference point, adjusted insufficiently from it, or justified acceptance by citing the AI tool's data access rather than team context [21][17, pp. 119–128]. *Algorithm aversion* was coded when a participant rejected a recommendation without evaluation or explicitly linked distrust to a prior AI tool error [10]. *Calibrated reliance* was coded when acceptance or rejection was explicitly grounded in named evidence about the recommendation's reliability in the current context [20].

Table 3.3 presents the complete coding evidence, showing the chain from participant utterance to raw code to final codebook code for all twelve codes. Each code is grounded in at least one verbatim transcript excerpt with timestamp.

Table 3.3: Complete coding evidence: transcript to codebook code for all twelve codes

Transcript excerpt	Raw code	Codebook code
P6, 9:54: "people don't process the AI suggest [sic] 42 as a data point. They process it as a reference"	number treated as reference not evidence	A1: Number-first contamination
P2, 14:36: "34 out of 42 is 80%. . . I would still shoot for to complete all 42"	reasons from AI figure not independently	A2: Reference-point adjustment
P7, 17:30: "I would not present the number before the team has done their own estimates"	deliberate withholding of AI figure	A3: Number withholding safeguard
P2, 17:59: "I am building a product and I need to be managing people"	sides with developer on relationship grounds without engaging AI evidence	B1: Developer authority assertion
P8, 15:27: "We got two signals pointing in different directions. Let us spend two minutes making sure we understand both before we move on"	facilitator requires both sources to articulate basis	B2: Examine both signals safeguard
P6, 13:27: "The AI pattern matching on similar stories can be deeply misleading. A story that looks similar on the surface may be touching a completely different infrastructure"	questions whether AI patterns apply to current context	B3: Pattern match scepticism

Transcript excerpt	Raw code	Codebook code
P4, 2:19: “Can you walk me through what you think makes this story different from the ones the tool is drawing on”	requests concrete basis before override	B4: Legitimate contextual dismissal
P6, 22:07: “The explanation needs to be a sentence, maybe two. That is enough. I do not need the model weights”	transparency of reasoning drives trust	C1: Transparency drives trust
P4, 23:48: “The tool was seeing something we didn’t give enough credit to... historical pattern captures better than individual intuitions”	trust updated by AI being correct when team dismissed it	C2: Track record drives trust
P9, 13:00: “We should definitely take the inputs from AI, but human should be the final one to decide”	final commitment must remain with team	C3: Human final authority
P6, 22:32: “The AI stays outside the planning room... It is a research tool for the Scrum Master and not a participant in the planning”	AI informs facilitator, facilitator decides what reaches team	C4: Facilitator as mediator
P8, 15:18: “I would check whether the AI was drawing on data from when the team was smaller and the codebase less stable”	trust calibrated by assessing whether AI patterns apply to current conditions	C5: Contextual applicability assessment

3.5 Model Construction and Validation

The conceptual model was built from the analysis through a three-phase construction process following Phase B HACO mapping.

In Phase B, the raw codes from Phase A were mapped onto HACO’s four meta-dimensions [12]. Task characteristics absorbed codes about what the AI tool should handle versus what humans should handle: number withholding codes, AI does analysis before session codes, and human makes final commitment codes all mapped here. Teaming characteristics absorbed codes about how the facilitator interacts with AI output and how that output reaches the team: facilitator reads privately codes, AI informs facilitation not team reasoning codes, and examine both signals codes all mapped here. The trust dimension absorbed codes about what makes practitioners trust or distrust AI tool recommendations: transparency of reasoning codes, track record codes, and contextual applicability codes all mapped here. The learning paradigm dimension absorbed codes about how practitioners update behaviour based on AI experience, but did not generate sufficient consistent themes to produce a model component, as documented in Section 5.

Table 3.4 shows the complete mapping of codebook codes to HACO dimensions and the model component each dimension produced.

Table 3.4: Phase B mapping: codebook codes to HACO meta-dimensions

HACO dimension	Codes mapped	Model component produced
Task characteristics	A1, A2, A3, C3	Role zone definitions (Zone 1, 2, 3) and number withholding principle
Teaming characteristics	B1, B2, B3, B4, C4	Interaction principles (examine both signals, override justification) and facilitator-as-mediator pattern
Trust	C1, C2, C5	Transparency principle and risk checklist trust-related conditions
Learning paradigm	Partial: C2 (human learning from experience)	No standalone component produced — insufficient consistent themes around AI tool adaptation

In Phase C, the HACO-mapped themes were used to construct the three model components. RQ1 findings identified where bias risks were highest. RQ2 findings identified what the model needed in terms of transparency requirements and human oversight. RQ3 is where those findings came together: we specified role definitions, interaction principles, and a risk checklist [12, 20]. The specific boundaries of each component, including the three zone definitions, the five interaction principles, and the eleven risk conditions, were determined entirely by the empirical data rather than prescribed in advance. The model was not designed before analysis; it emerged from the thematic findings.

The three-component structure of the model, role zone definitions, interaction principles, and a risk identification checklist, was not derived from HACO directly. HACO provided the analytical lens for organising themes but does not prescribe what a sprint planning collaboration model should look like. The zone structure emerged because participants consistently described three empirically distinct layers of AI involvement: pre-session data work with no team contact, facilitator-mediated interpretation, and human-only final decisions. The interaction principles emerged because participants described specific deliberate practices they follow when working with AI recommendations, practices that were recurring and independently described across multiple participants. The risk checklist emerged because participants described specific conditions under which bias risk increases, conditions concrete enough to be actionable. None of these three components exist in HACO. They are the empirical contribution that emerged from applying HACO as a coding lens to sprint planning practitioner data.

Structured walkthroughs were conducted with three of the original participants prior to TD1 submission to assess the model's practical credibility. Participants were selected purposively to include experienced Scrum Masters who had given the richest interview data: P6, P7, and P8. Each participant reviewed the model figure and responded to three open-ended questions covering the zone definitions, interaction principles, and risk checklist. This form of assessment does not establish predictive or statistical validity, but evaluates whether the model's components are plausible and consistent with practitioner experience. Components receiving consistent critical feedback across multiple participants were revised; contradictory or isolated feedback was documented in the Discussion chapter.

3.6 Validity and Reliability

Several threats to validity were considered. *Researcher bias in coding* was mitigated through a systematic coding process in which operational definitions were explicitly defined and refined before being applied consistently across all nine transcripts [9, pp. 251–252]. Where code boundaries were ambiguous, definitions were sharpened through deliberate reflection to ensure consistent application. *Participant self-reporting* was partially mitigated by the vignette approach, though it does not eliminate the issue [1, 23].

Sprint planning is a team event, but we interviewed individuals. The model is grounded in individual practitioner experience, not an empirically validated team-level process. Individual mental models are where bias begins [17, pp. 20–24], but team-level validation remains as future work.

Generative AI tools were used for language assistance including grammar correction and sentence refinement, to explore and discuss concepts during the research design phase, and to aid comprehension of methodological approaches. The research design, literature selection, data collection, analysis, interpretation, and all written arguments are our own. All cited references were read and verified by the authors. GenAI use was declared to and discussed with the supervising instructor throughout the project, in accordance with the BTH GenAI policy.

Chapter 4

Results and Analysis

4.1 Overview of Participants

Nine Agile practitioners participated in the study, representing a range of roles, experience levels, and AI tool familiarity (Table 3.2). Roles included Scrum Masters (P3, P4, P6, P7, P8), an engineering manager (P1), tech leads (P5, P9), and a team lead (P2). Experience ranged from approximately five to twenty-one years. Five participants had direct experience with AI or data-driven planning tools. Two had partial experience, primarily AI coding assistants. Two had no AI planning tool experience but had discussed adoption within their organisations.

This variation was intentional. Practitioners without AI tool experience provided useful perspective on the conditions under which adoption is resisted, which directly informed RQ2.

4.2 RQ1a: Effects of Anchoring Bias and Algorithm Aversion on Decision Quality

4.2.1 Anchoring Bias

Anchoring bias was the most consistently and independently identified cognitive risk across the study. Five of nine participants described the mechanism of number-first contamination, the risk that an AI-suggested figure shared before team estimation would shape all subsequent reasoning, without being prompted to name it as a bias.

P6 gave the clearest articulation: *"People don't process the AI suggest [sic] 42 as a data point. They process it as a reference."* P3: *"Once a specific number is out there, it is like very hard for people to think past that."* P4: *"I'm aware from the experience that putting a number in the room early shapes the conversation in ways that are hard to undo."*

Two participants described first-hand experience of this mechanism in real planning sessions. P7: *"This scenario is almost exactly of what we had lived through, which is a bit eerie."* P8: *"That's something I learned earlier on by trying another way."* Both subsequently changed their facilitation practice as a result.

P2, the participant with the least AI tool experience, showed the clearest anchoring response in the vignette: *"34 out of 42 is 80%... I feel comfortable just taking on... I would still shoot for to complete all 42."* The AI figure became the starting point from which reasoning adjusted, rather than an independent judgement being formed first.

The anchoring risk is highest when the AI tool's figure is shared before independent team estimation and when the team lacks an experienced facilitator who recognises and actively mitigates the risk.

4.2.2 Algorithm Aversion

Responses to Vignette B revealed patterns related to algorithm aversion [10]. As noted in the methodology, the scenario operationalises the dismissal-without-engagement pattern associated with algorithm aversion risk, rather than the strict experimental condition of prior observed algorithm error. With this in mind, rather than outright rejection, experienced practitioners showed calibrated scepticism, questioning the AI tool's pattern-matching basis before accepting or rejecting a recommendation.

P6: *"The AI pattern matching on similar stories can be deeply misleading. A story that looks similar on the surface may be touching a completely different infrastructure."* This is a legitimate challenge to the AI tool's evidence base, not straightforward aversion.

P2 showed the clearest aversion pattern: siding with the developer on relationship grounds without engaging with the AI tool's evidence. *"The goal is not to get the story completed in this sprint. I am building a product and I need to be managing people."*

P4 provided the most instructive example: the team sided with the developer against an AI tool warning, and the story ran approximately 45% over estimate. *"The tool was seeing something we didn't give enough credit to... The developer knew the code but was underestimating the testing complexity, which historical pattern captures better than individual intuitions."* This experience subsequently changed P4's facilitation approach.

The key distinction that emerged is between legitimate dismissal with concrete evidence that the AI tool's patterns do not apply, and unjustified dismissal through assertion of domain knowledge without articulating a specific basis.

4.3 RQ1b: How Cognitive Biases Manifest in Practitioner Reasoning

4.3.1 Anchoring manifestation

Anchoring manifested in two observable patterns. The first is *reference-point adjustment*: participants who encountered the number 42 began reasoning from it and adjusted downward rather than forming an independent judgement. P2's response was the clearest instance.

The second is *number withholding as a learned safeguard*: five of nine participants described a deliberate practice of keeping the AI tool's figure private until after team estimation. P8: *"I note the 42 privately, run the session normally and see where the team lands."* P7: *"I would not present the number before the team has done their own estimates."* P3: *"I would look at it privately before the session."*

The number-withholding safeguard is one of the clearest empirical findings of this study. It strengthens the model's empirical grounding because it emerged from practitioner experience rather than from researcher prescription. Five participants arrived at the same practice independently, two of them explicitly after experiencing anchoring effects in real sessions.

4.3.2 Algorithm aversion manifestation

Algorithm aversion manifested primarily as *developer authority assertion*: invoking domain expertise to dismiss an AI tool flag without articulating a specific basis for doing so.

Experienced facilitators described challenging this pattern directly. P8: *"I still want them to articulate the specific differences rather than assert it."* P4: *"Can you walk me through what you think makes this story different from the ones the tool is drawing on?"* P7: *"My job is to make sure that the team has examined both inputs properly and not to decide which is right."*

A secondary manifestation was *pattern match scepticism*: questioning whether the AI tool's historical patterns apply to the current context. P8 described checking whether the AI tool was drawing on data from when the team was smaller and the codebase less stable. This is calibrated scepticism that engages with the evidence rather than dismissing it outright.

4.4 RQ2: Trust Calibration in AI-Assisted Sprint Planning

4.4.1 Conditions driving trust

Across the interviews, *transparency of reasoning* was the most consistently identified trust condition: AI tool recommendations that explained their basis were trusted more than those providing output only. P6 described transparency as non-negotiable; P7 called it table stakes.

P6 was specific about what level of explanation was sufficient: *"The explanation needs to be a sentence, maybe two. Based on your last eight similar stories, this one overran by 15 percent. That is enough. I do not need the model weights."* P3: *"Maybe if it could show me specifically which past stories it is drawing on and what actually went wrong with them."* P2: *"It exposes if there are some biases and assumptions baked in there. You could just correct or update those assumptions."*

Transparency enables calibration. Practitioners can only assess whether an AI tool's evidence applies to their specific context if they can see what that evidence is.

Track record over time was the second trust driver. P4's experience of the AI tool being right when the team dismissed it directly changed facilitation practice. Trust is not static; it is updated by experience.

Contextual applicability was the third driver. Participants consistently assessed whether the AI tool's historical patterns applied to current conditions before deciding how much weight to give a recommendation.

4.4.2 Conditions driving distrust

Distrust was most commonly triggered by three conditions: recommendations without explanation, perceived inapplicability of historical patterns to the current context, and lack of prior experience with the tool. P9, in the early stages of AI tool adoption, described needing to build trust through previous outcomes before acting on recommendations.

4.4.3 The facilitator as trust mediator

A recurring pattern across the interviews was that practitioners did not want the AI tool to speak directly to the team before estimation. Five of nine participants independently described the same setup: the AI tool informs the facilitator before the session, and the facilitator decides how, or whether, to introduce that information into the planning discussion.

P6: *"The AI stays outside the planning room. It does its work beforehand, surfaces patterns, flags risks, gives me context, and then I decide what is relevant enough to bring to the session and how to frame it. It is a research tool for the Scrum Master and not a participant in the planning."* P8: *"The tool should be informing my facilitation, not shaping the team's reasoning directly."*

This model resolves the anchoring risk structurally. If the AI tool's figure never enters the room directly, it cannot contaminate team estimation. The facilitator filters AI tool output through contextual knowledge the tool lacks, and decides whether the signal is worth raising and how to frame it so it informs rather than anchors.

4.5 RQ3: The Conceptual Model

The conceptual model was constructed from the thematic findings of RQ1 and RQ2. It specifies three components: role definitions, interaction principles, and a risk identification checklist.

4.5.1 Component 1: Role Definitions

Zone 1, AI analytical support before the planning session: Tasks that are data-intensive, do not require contextual knowledge the AI tool lacks, and do not involve direct influence on team reasoning. These include historical velocity trend analysis, pattern recognition across similar past stories, risk flagging based on historical overruns, capacity baseline calculation, and preliminary story complexity flagging from historical effort data.

Zone 2, Human-AI dialogue facilitated by the Scrum Master: Tasks where the AI tool provides data and the facilitator provides context. AI tool output is not shared directly with the team but used by the facilitator to structure discussion. These include interpreting AI tool risk flags, assessing whether historical patterns apply to current stories, story prioritisation where the AI tool provides data and the facilitator frames the question, and post-estimation comparison.

Zone 3, Human only, final decisions: Tasks requiring knowledge the AI tool cannot access or carrying accountability implications that must remain with humans. These include the final sprint commitment, override decisions with explicit justification, individual performance and capacity judgements, and decisions requiring knowledge of recent changes not reflected in the AI tool's data. P8: *"Judgement does not come from the tool. It comes from knowing the team, knowing the context, and being present in the room in a way the tool never is."*

4.5.2 Component 2: Interaction Principles

Principle 1, number withholding before independent estimation: AI tool-suggested figures are reviewed by the facilitator privately. Specific figures are not shared with the team before independent estimation. The team estimates first. The AI tool's figure is used only as a post-estimation sanity check. Five of nine participants described this as deliberate practice; two developed it in direct response to experiencing anchoring effects.

Principle 2, reasoning transparency requirement: AI tool recommendations must include their basis, which specific past stories, what patterns, what went wrong. Transparency of reasoning was consistently identified as a key trust condition across participants. P6 described it as non-negotiable; P7 called it table stakes. The required level is modest: one or two sentences, not model weights.

Principle 3, examine both signals: When the AI tool and a developer disagree, neither is dismissed without examination. The facilitator asks both sources to articulate their basis. P8: *"We got two signals pointing in different directions. Let us spend two minutes making sure we understand both before we move on."* This principle is supported across codes B2 and B4 combined. While B2 (examine both signals safeguard) fires explicitly in P8, the facilitation behaviour of requiring articulated basis before deciding also appears in P4 and P7 under B4 (legitimate contextual dismissal): P4 asked developers to walk through what makes a story different from the AI tool's historical patterns, and P7 described her role as ensuring the team has examined both inputs properly. Across B2 and B4 combined, four of nine participants described this facilitation practice.

Principle 4, explicit override justification: When the team overrides an AI tool recommendation, the reason must be stated concretely. Not "the tool does not know our codebase" but "the tool is drawing on stories from two years ago before we refactored the legacy module."

Principle 5, human final authority: The final sprint commitment is always made by the team. The AI tool informs; humans decide. This was stated or implied by all nine participants. P9: *"We should definitely take the inputs from AI, but human should be the final one to decide."*

4.5.3 Component 3: Risk Identification Checklist

High risk conditions:

1. AI tool recommendation presented before team independent estimation: apply Principle 1 (P3, P4, P7, P8)

2. First sprint using a new AI planning tool: calibrated trust not yet established (P1, P3, P9)

3. High pressure release cycle: heuristic reasoning more likely, apply Principle 3 (P1, P3, P4, P6, P7, P8)
4. Senior team member's judgement goes unchallenged: anchoring and groupthink risk, apply Principle 3 (P1, P3, P6, P7)
5. AI tool recommendation with no explanation of basis: apply Principle 2 (P2, P4, P6, P7, P8)

Moderate risk conditions:

6. Participants unfamiliar with AI tool data sources (P3, P4, P6, P8, P9)
7. Significant codebase changes since AI tool training data: apply Principle 4 (P3, P8)
8. Long gap since similar work was last done (P8)
9. Developer confidence without articulated basis: apply Principles 3 and 4 (P3, P4, P5, P6, P7, P8)
10. Significant team composition change since AI tool training data: apply Principle 4 (P3, P7, P9)
11. Sprint inertia: previous good sprint run creating collective overconfidence: apply Principle 3 (P7, P8)

Figure 4.1 provides a figurative representation of the model, illustrating the relationships between the three components and the flow of information between zones. The figure is intended as a visual aid to support reading of the textual model specification; the authoritative description of each component is provided in the sections above and summarised in Table 4.1.

4.5.4 How the Model Answers the Research Questions

The three components of the model map directly onto the four research questions that guided this study. RQ1a asked how anchoring bias and algorithm aversion affect decision quality in AI-assisted sprint planning. The model answers this through the risk identification checklist, which specifies eleven conditions under which anchoring and algorithm aversion are most likely to emerge. Five of the eleven conditions directly address anchoring risk, two address algorithm aversion risk, and the remaining four address conditions that amplify both. The checklist gives teams a practical diagnostic tool for identifying when bias risk is elevated before a planning session begins.

RQ1b asked how these biases manifest in practitioner reasoning. The model answers this through the codebook and the interaction principles. Anchoring manifests as reference-point adjustment, which Principle 1 (number withholding) directly prevents by structurally separating AI output from team estimation. Algorithm aversion manifests as developer authority assertion, which Principle 3 (examine both signals)

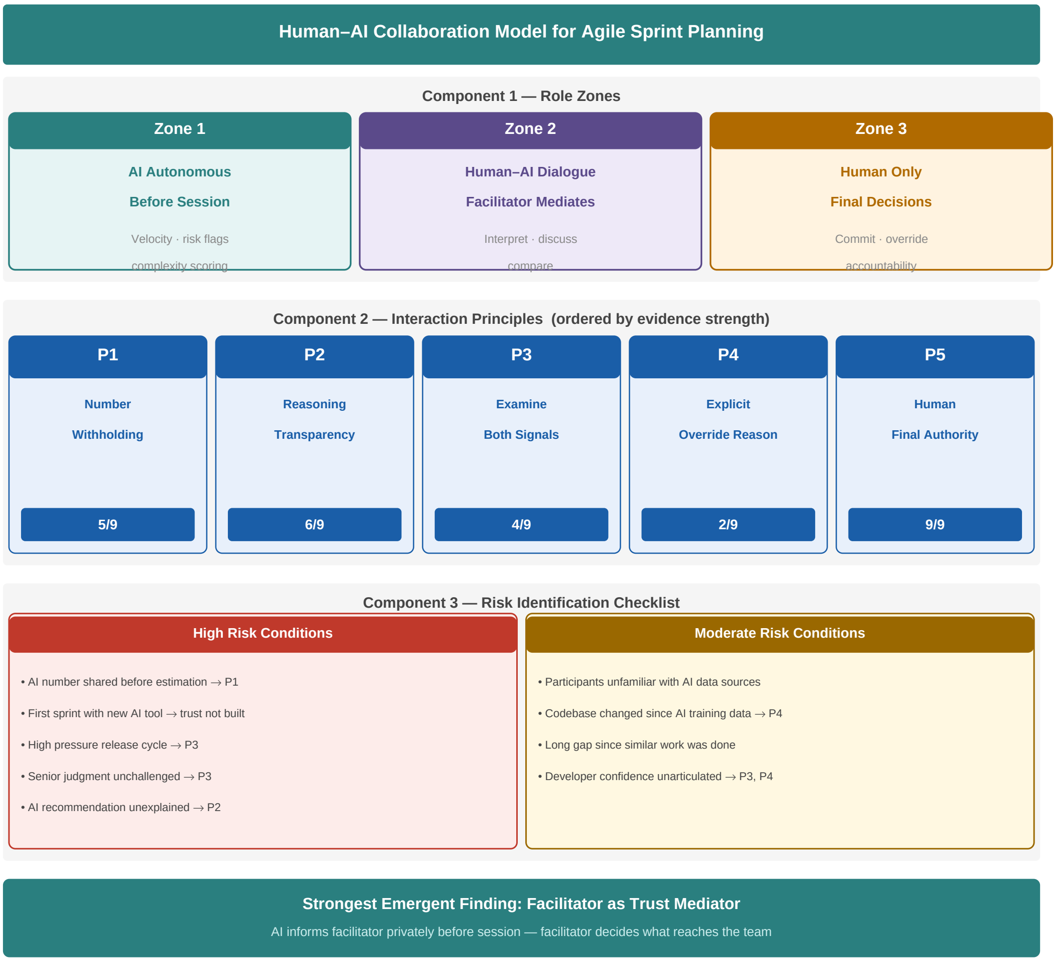

Figure 4.1: Figurative representation of the Human-AI collaboration model for Agile sprint planning

Table 4.1: Summary of model components

Component	Description
Role Zone 1	AI analytical support before session: velocity analysis, risk flagging, preliminary complexity flagging
Role Zone 2	Human-AI dialogue: facilitator interprets AI tool signals, post-estimation comparison
Role Zone 3	Human only: final commitment, override decisions, performance judgement
Principle 1	Withhold AI tool figure until after independent team estimation
Principle 2	Require AI tool to explain reasoning basis
Principle 3	Examine both AI tool and developer signals before deciding
Principle 4	Require explicit concrete justification for any override
Principle 5	Human final authority on all commitment decisions
Risk checklist	5 high-risk and 6 moderate-risk conditions with mapped mitigations

and Principle 4 (explicit override justification) directly counter by requiring articulated evidence before either signal is dismissed.

RQ2 asked how practitioners calibrate trust in AI tools. The model answers this through Zone 2 and Principle 2. Zone 2 positions the facilitator as the trust mediator between AI output and team reasoning, creating the conditions for calibrated engagement rather than either blind acceptance or blanket rejection. Principle 2 requires the AI tool to explain its reasoning basis, which is the precondition for calibration identified by Lee and See [20]: practitioners cannot calibrate reliance to actual capability if they cannot see the basis for a recommendation.

RQ3 asked what principles should govern Human-AI collaboration in sprint planning. The model is the direct answer: five interaction principles, three role zones, and eleven risk conditions derived from practitioner experience and grounded in the HACO taxonomy and Lee and See's calibrated trust theory.

4.5.5 Practical Application in Sprint Planning

The model is designed to be used by a Scrum Master or sprint planning facilitator. The following describes how the three components work together in a real planning session.

Before the session. The Scrum Master reviews the AI tool's output privately, without sharing it with the team. Using the risk checklist, they assess whether any high-risk or moderate-risk conditions apply: Is this the first sprint with a new AI tool? Is there a high-pressure release cycle? Has the codebase changed significantly since the AI tool's training data? Based on this assessment, they decide which AI signals are worth raising during the session and how to frame them. The AI tool's specific numerical recommendations are not brought into the room at this stage.

During the session. The team estimates independently using their usual process, without any AI figure in the room. This preserves independent team estimation and prevents number-first contamination. Only after the team has committed to their estimates does the facilitator introduce any relevant AI signals, framing them as data points to consider rather than recommendations to accept or reject. If the AI tool flagged a specific story as high risk, the facilitator uses Principle 3 (examine both signals) to surface the disagreement and ask the assigned developer to articulate why the story differs from the historical patterns the AI tool is drawing on. If the team decides to override the AI tool's signal, Principle 4 requires them to state a concrete reason, not just assert domain knowledge.

After the session. The facilitator compares the team's committed sprint scope against the AI tool's recommendations and notes any significant gaps. Over time, tracking whether the AI tool was right in cases where the team overrode it builds the track record that enables calibrated trust. This is how trust updates from experience rather than from a priori assumption.

The model does not prescribe every facilitation decision. It sets the structural conditions under which Human-AI collaboration is less likely to produce anchoring or algorithm aversion, while leaving facilitation judgement where it belongs: with the experienced practitioner in the room.

Chapter 5

Discussion

5.1 Interpretation of Findings

5.1.1 Anchoring bias in AI-assisted sprint planning

Prior work has documented anchoring as a general pattern in software estimation [21]. What this study adds is grounding at the specific moment of interaction: the point at which an AI tool's figure enters a planning room. The recommendation arrives before independent team estimation, becomes the first concrete number in the room, and shapes all subsequent reasoning as a reference point rather than a data point. The distinction, a figure treated as a reference versus a figure treated as evidence, was articulated most precisely by P6 and is the practical crux of the anchoring problem in AI-assisted planning.

The number-withholding safeguard is one of the clearest empirical findings of this study. It emerged independently as a learned practice from practitioners who had experienced anchoring effects first-hand, not as a recommendation suggested by the researchers. This makes it stronger evidence than a theoretically derived prescription would be.

The finding also connects to Drury-Grogan and O'Dwyer's [11] observation that time pressure increases heuristic reasoning. When a number is already in the room, time-pressured teams are more likely to adjust from it rather than form independent judgements. An AI tool recommendation arriving before estimation therefore combines two bias-amplifying conditions at the same moment.

5.1.2 Algorithm aversion in context

The form of algorithm aversion observed here is more nuanced than Dietvorst et al.'s [10] original characterisation. Most participants did not show blanket rejection but contextual scepticism, questioning whether the AI tool's historical pattern was applicable to the current situation before deciding how much weight to give it.

This likely reflects the specific population. All participants were experienced practitioners with more than five years in Agile roles. Experienced practitioners may have developed more sophisticated approaches to evaluating AI recommendations precisely because they have enough domain knowledge to interrogate the tool's evidence base.

The most concerning instance of algorithm aversion came from P2, who sided with the developer on relationship grounds without engaging with the AI tool's evidence at all. P2 also had the least AI tool experience of all participants. This is consistent with the idea that aversion is strongest when trust has not yet been calibrated through experience.

The example from P4, where the AI tool was right and the team dismissed it with the story running 45% over estimate, illustrates both the cost of algorithm aversion and the fact that it can be corrected through experience. Organisations introducing AI planning tools should create conditions for practitioners to reflect on outcomes and update their trust calibration over time.

5.1.3 Trust calibration

The consistent identification of transparency of reasoning as a key trust condition connects directly to Lee and See's [20] definition of appropriate reliance as trust proportional to a tool's actual capabilities. Practitioners cannot calibrate reliance to actual capability if they cannot see the basis for a recommendation. Transparency is the precondition for calibration.

The level of transparency participants required was notably modest: one or two sentences. The barrier to enabling calibrated trust through transparency is low. The design requirement is not complex explainability infrastructure but minimal traceability.

The facilitator-as-mediator pattern connects to Emmanouilidis et al.'s [14] observation that interaction protocol design, not AI tool accuracy, determines whether collaboration produces better outcomes. The number-withholding practice is exactly such a protocol: a deliberate choice about when and how AI tool output enters the planning room. This study provides empirical evidence that this protocol matters and that experienced practitioners have independently converged on it.

5.1.4 Implications for AI tool design

A broader implication of these findings is that many current AI-powered sprint planning tools may be designed in ways that inadvertently increase cognitive bias risk. By presenting numerical recommendations upfront, such tools introduce anchoring at the earliest stage of decision-making. The findings suggest that the primary limitation of existing tools is not necessarily predictive accuracy, but the interaction design through which recommendations reach the team. Even a highly accurate tool may degrade decision quality if its output enters the planning room before independent estimation has taken place. This suggests that improving AI accuracy alone is insufficient; without appropriate interaction design, even correct recommendations may systematically degrade team decision quality. Tools that surface recommendations to the facilitator first, and provide clear reasoning behind them, are likely to produce better collaborative outcomes than tools that simply output a figure for the team to react to.

5.2 The Conceptual Model in Context

5.2.1 How the model relates to HACO

The conceptual model builds on the HACO taxonomy [12] but differs from it in important ways. HACO organises Human-AI teaming across four meta-dimensions: task characteristics, learning paradigm, teaming characteristics, and trust. All four were used as a coding lens during Phase B of the thematic analysis. Three of the four produced model components with sufficient empirical grounding.

Task characteristics shaped the role zone definitions, determining which planning tasks belong to the AI tool, which require Human-AI dialogue, and which must stay under human control. Teaming characteristics fed into the interaction principles, particularly the facilitator-as-mediator pattern and the examine-both-signals principle. The trust dimension informed the transparency requirement, the override justification principle, and the risk checklist conditions.

The learning paradigm dimension, which captures how both human and AI learn from the interaction over time, did not produce a standalone model component. Participants described human learning from experience — P4's account of updating facilitation practice after the AI tool proved correct is the clearest example, and P7 and P8 both described changing their facilitation approach after experiencing anchoring effects firsthand. However, the data did not surface consistent themes around how AI tools themselves learn or adapt within sprint planning contexts.

Three factors likely explain this absence. First, the interview format captures a single point in time. Practitioners can describe how they have changed their behaviour over multiple sprints, but observing AI tool adaptation requires longitudinal data that a one-time interview cannot provide. Second, current AI planning tools, including the LinearB tool used by P4, do not visibly adapt their recommendations based on team feedback in ways practitioners notice or discuss. The learning happens invisibly in model retraining cycles that are opaque to the user. Third, the interview guide did not include a direct question about AI tool adaptation, which may have limited how much this theme was surfaced. Whether richer probing on this dimension, or a longitudinal study design, would produce stronger data is an open question we flag as a direction for future investigation.

The model that emerged goes beyond what HACO alone could provide because it is grounded in empirical data from a specific context. HACO identifies calibrated trust as a desirable state but does not specify what causes practitioners to drift toward anchoring or algorithm aversion, or what safeguards prevent it. Our model fills that gap.

5.2.2 Relationship to Lee and See's trust theory

Lee and See's [20] definition of calibrated trust provides the theoretical foundation for the model's trust-related principles. Principle 2, the reasoning transparency requirement, directly operationalises calibrated trust. Principle 1, number withholding, addresses a condition that Lee and See's theory implies but does not specify: when an AI tool recommendation enters the room before independent estimation, the conditions for calibrated trust are undermined regardless of how transparent the tool is.

Number withholding creates the precondition for calibrated trust by ensuring team reasoning forms independently before AI tool output is introduced.

5.2.3 Validation findings

Structured walkthroughs with P6, P7, and P8 assessed the model's practical credibility prior to submission. All three participants confirmed the zone structure as realistic and consistent with their own facilitation practice. P6 described Zone 2 as reflecting exactly how they already work: the AI does its background work before the session, the facilitator decides what to bring in, and the team calls the commitment. P7 noted that the boundary between Zone 1 and Zone 2 is the critical design decision, having experienced firsthand how AI output entering the room directly contaminated team estimation during a pilot with a sprint planning tool. P8 described Zone 2 as the AI being *"background, not foreground"*: a *"well-prepared assistant who has done the background research"*, not a co-facilitator whose recommendations need to be addressed in real time.

On the interaction principles, all three confirmed number withholding as the most important and most immediately applicable. P7 described an experience where AI output reached the team before independent estimation had taken place, resulting in a sprint the team later described as a disaster: they finished at 26 points, one of their worst ever, reinforcing why number withholding matters in practice. P8 added that override justification should be framed positively: being explicit about why a decision differs from the AI output, rather than treating the AI recommendation as a default that needs defeating.

Two refinements were made following the walkthroughs. First, Zone 1 now explicitly includes flagging stories where the backlog is not groomed enough to estimate, a pre-session signal P6 identified as practically useful that the original model did not cover. Second, two conditions were added to the risk checklist: sprint inertia, where previous good sprints create collective overconfidence that is distinct from general high-pressure conditions, raised by P7; and significant team composition change since the AI tool's training data, raised by P8 from a direct experience where velocity data from a larger team produced misleading recommendations.

Components that received isolated or contradictory feedback were documented rather than revised. P7 suggested making the timing of facilitator disclosure during the session explicit in Principle 1. This was noted but not incorporated, as the principle intentionally leaves facilitation sequencing to the facilitator's judgement.

5.2.4 Scope and generalisability

The model applies to Agile sprint planning. Whether the principles generalise to other Agile ceremonies is an empirical question this study cannot answer. Some principles, particularly human final authority and reasoning transparency, are likely relevant in any Human-AI decision-making context. Others, particularly number withholding, are specific to estimation contexts where a figure can become an anchor.

The model is also scoped to the individual-interview level. The next step is to test whether applying the model at the facilitation level produces measurable effects at the team level.

5.3 Limitations

This study produced a conceptual model, not a deployable tool or a statistically generalisable theory. The study is limited by its small sample size (n=9), which constrains generalisability beyond the studied context. Nine participants is consistent with the point at which thematic saturation was reached: no substantially new codes emerged after the seventh interview, and the final two interviews primarily reinforced existing themes. P8 reinforced the anchoring and facilitator-as-mediator findings, while P9, approaching AI adoption from an early-stage perspective, confirmed the trust calibration pattern without introducing new codes. The findings should therefore be interpreted as exploratory rather than representative of the broader Agile practitioner population [9, p. 146].

The model is scoped to sprint planning. The two participants without AI tool experience provided useful perspective on non-adoption, but their vignette responses were necessarily hypothetical rather than grounded in direct experience with such tools.

The model's practical credibility was assessed through structured walkthroughs with three original participants prior to TD1 submission. The walkthroughs confirmed the model's plausibility and produced two refinements incorporated into this version. The model's generalisability beyond the studied context remains a limitation regardless of this assessment.

5.4 Ethical and Societal Considerations

The most significant ethical concern is *deskilling*. As AI tools take on more analytical work in sprint planning, practitioners risk losing competence in independent reasoning. Research on automation has documented this pattern [24]. The model includes explicit checkpoints, the examine-both-signals principle and the explicit override justification requirement, that require active human engagement rather than passive acceptance.

All participant identities and organisations were fully anonymised. No sprint records, backlog items, or organisational data were collected. Research instruments, the anonymised codebook, and data documentation will be published as a Zenodo package prior to final thesis submission. Full interview transcripts will not be publicly released due to participant confidentiality commitments, but are available to the thesis supervisor and examiner upon request.

Chapter 6

Conclusions and Future Work

6.1 Answers to Research Questions

RQ1a: How do anchoring bias and algorithm aversion affect decision quality in Agile sprint planning when AI recommendations are introduced?

Among the two bias mechanisms examined, anchoring appeared more consistently and more severely across the study. Five of nine participants independently described number-first contamination as a real and recurring risk, with two reporting direct first-hand experience of it affecting real planning sessions. Algorithm aversion appeared in a more nuanced form than prior literature describes: experienced practitioners showed calibrated scepticism rather than blanket rejection, distinguishing between legitimate dismissal based on identifiable contextual differences and unjustified dismissal through assertion of domain authority.

RQ1b: How do these biases manifest in practitioner reasoning during AI-assisted sprint planning?

Anchoring manifests through reference-point adjustment: when an AI tool figure is shared before independent estimation, team reasoning adjusts from it rather than forming an independent judgement. Algorithm aversion manifests through developer authority assertion, invoking domain expertise to dismiss an AI tool flag without articulating a specific basis. Experienced facilitators have developed explicit practices to counter both: number withholding for anchoring and examine-both-signals facilitation for algorithm aversion.

RQ2: How do Agile practitioners perceive and calibrate their trust in AI tools when delegating planning tasks, and what conditions lead to overtrust or distrust?

Trust calibration is shaped by three conditions: transparency of AI tool reasoning, track record over time, and contextual applicability of historical patterns. A recurring pattern across the interviews was that experienced practitioners independently converged on a facilitator-as-mediator model, in which the AI tool informs the facilitator privately before the session and the facilitator decides what to bring to the team and how to frame it. This model protects independent team estimation while preserving access to the AI tool's pattern-recognition capability.

RQ3: What principles should govern Human-AI collaboration in Agile sprint planning to reduce cognitive bias risk while maintaining human oversight and team autonomy?

The conceptual model specifies five interaction principles: (1) withhold AI tool-suggested figures until after independent team estimation; (2) require the AI tool to explain the basis for its recommendations; (3) examine both AI tool and developer signals before deciding; (4) require explicit concrete justification for any override; and (5) maintain human final authority on all commitment decisions. These are accompanied by three-zone role definitions and an eleven-condition risk identification checklist.

6.2 Contributions

This thesis makes two contributions to the Human-AI collaboration literature in software engineering.

The first is an **empirical contribution**: interview data from nine Agile practitioners documenting how anchoring bias and algorithm aversion manifest in reasoning about AI tool recommendations during sprint planning, and how trust in AI tools develops and is calibrated in that context. Based on the reviewed literature, no prior study was found combining practitioner experience data on cognitive bias and trust in AI-assisted sprint planning in a single study. The interview guide, consent form, anonymised codebook, and data documentation will be made available as a Zenodo package. Full transcripts will not be publicly released due to participant confidentiality.

The second is the **conceptual model**: a structured specification of role definitions, interaction principles, and risk conditions for Human-AI collaboration in Agile sprint planning, grounded in practitioner experience. The facilitator-as-mediator pattern is an original empirical finding of this study. It was not derived from HACO, Lee and See's trust theory, or any prior literature: it emerged independently across five participants who described the same interaction model without being prompted, making it stronger evidence than a theoretically derived recommendation would be. This pattern is the core practical contribution of the model — a structural safeguard against anchoring bias that does not depend on individual practitioner vigilance but on how the interaction between AI tool and team is designed. The model as a whole is a design artefact that practitioners and researchers can use as a reference when structuring Human-AI collaboration in sprint planning contexts.

6.3 Future Work

The model is grounded in individual practitioner experience and has not been tested at the team level. Testing the model in real sprint planning sessions with full teams is the natural next step. A study comparing planning sessions with and without the interaction principles applied would provide evidence of whether the principles reduce anchoring and algorithm aversion at the team level.

Future work could extend the model to other Agile ceremonies and investigate whether the same bias patterns appear in other AI-assisted decision-making contexts in software engineering.

The facilitator-as-mediator finding raises a practical question this study could not answer: what happens in teams without a dedicated Scrum Master? The model's effectiveness depends on facilitation competence, which may limit its applicability in flat or self-organising teams where no single person holds that role. Investigating how the principles can be operationalised without a dedicated facilitator is a relevant extension.

The learning paradigm dimension of HACO did not produce a standalone model component in this study, as the data did not surface consistent themes around AI tool adaptation over time. Whether this reflects a genuine absence in current tools, a limitation of the sample, or a phenomenon that requires longer-term observation to capture is an open question. Future work investigating how AI planning tools learn from team behaviour across sprints, and how practitioners respond to that adaptation, would address this gap directly.

Finally, the transparency finding, that practitioners need only one or two sentences of explanation to enable calibrated trust, has direct implications for AI tool design. Future work could investigate whether building minimal traceability into AI planning tools measurably improves trust calibration and reduces both anchoring and algorithm aversion in practice.

Bibliography

[1] H. Aguinis and K. J. Bradley, "Best practice recommendations for designing and implementing experimental vignette methodology studies," *Organizational research methods*, vol. 17, no. 4, pp. 351–371, 2014.

[2] F. Aman, S. R. A. Hamid, and A. M. Isa, "A systematic review of trends in human–AI collaboration research," *International Journal of Technology, Knowledge & Society: Annual Review*, vol. 21, no. 1, 2025.

[3] J. Angara, S. Prasad, and G. Sridevi, "DevOps project management tools for sprint planning, estimation and execution maturity," *Cybernetics and Information Technologies*, vol. 20, no. 2, pp. 79–92, 2020.

[4] Y. Bao, X. Cheng, T. De Vreede, and G.-J. De Vreede, "Investigating the relationship between ai and trust in human-ai collaboration," in *Proceedings of the 54th Hawaii International Conference on System Sciences*, 2021.

[5] C. Barua, J. U. Z. Kabir, K. R. Alam, A. R. Nabil, and S. Duman, "AI-augmented agile project management in engineering: A framework for smart decision-making and risk mitigation," *International Journal of Science and Research Archives*, vol. 15, no. 3, pp. 967–973, 2025.

[6] K. Borowa, S. Kamoda, P. Ogrodnik, and A. Zalewski, "Fixations in agile software development teams," *Foundations of Computing and Decision Sciences*, vol. 48, no. 1, pp. 3–18, 2023.

[7] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative research in psychology*, vol. 3, no. 2, pp. 77–101, 2006.

[8] M. Choetkiertikul, H. K. Dam, T. Tran, T. Pham, A. Ghose, and T. Menzies, "A deep learning model for estimating story points," *IEEE Transactions on Software Engineering*, vol. 45, no. 7, pp. 637–656, 2018.

[9] J. W. Creswell, *Qualitative Inquiry and Research Design: Choosing Among Five Approaches*, 3rd ed. Thousand Oaks, CA: SAGE Publications, 2013.

[10] B. J. Dietvorst, J. P. Simmons, and C. Massey, "Algorithm aversion: people erroneously avoid algorithms after seeing them err," *Journal of experimental psychology: General*, vol. 144, no. 1, pp. 114–126, 2015.

[11] M. L. Drury-Grogan and O. O'Dwyer, "An investigation of the decision-making process in agile teams," *International Journal of Information Technology & Decision Making*, vol. 12, no. 06, pp. 1097–1120, 2013.

[12] A. Dubey, K. Abhinav, S. Jain, V. Arora, and A. Puttaveerana, "HACO: a framework for developing human-AI teaming," in *Proceedings of the 13th Innovations in Software Engineering Conference*, 2020, pp. 1–9.

[13] T. Dybå and T. Dingsøyr, "Empirical studies of agile software development: A systematic review," *Information and software technology*, vol. 50, no. 9-10, pp. 833–859, 2008.

[14] C. Emmanouilidis, J. Zotelli, K. Hengel, S. Waschull, and J. B. Bokhorst, "Co-creation design patterns for human-AI teaming in manufacturing and multi-domain decision-making," *IFAC-PapersOnLine*, vol. 59, no. 24, pp. 227–232, 2025.

[15] M. Hamza, D. Siemon, M. A. Akbar, and T. Rahman, "Human-AI collaboration in software engineering: Lessons learned from a hands-on workshop," in *Proceedings of the 7th ACM/IEEE International Workshop on Software-intensive Business*, 2024, pp. 7–14.

[16] M. F. Hamza, R. B. AbuRass, M. B. AbuRass, and T. B. Jber, "Agile project management optimization with AI technology: Challenges and impact," in *2025 12th International Conference on Information Technology (ICIT)*. IEEE, 2025, pp. 393–397.

[17] D. Kahneman, *Thinking, Fast and Slow.* New York: Farrar, Straus and Giroux, 2011.

[18] B. Klieger, C. Charitsis, M. Suzara, S. Wang, N. Haber, and J. C. Mitchell, "ChatCollab: Exploring collaboration between humans and AI agents in software teams," 2024.

[19] E. Kula, A. Van Deursen, and G. Gousios, "Context-aware automated sprint plan generation for agile software development," in *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering*, 2024, pp. 1745–1756.

[20] J. D. Lee and K. A. See, "Trust in automation: Designing for appropriate reliance," *Human factors*, vol. 46, no. 1, pp. 50–80, 2004.

[21] R. Mohanani, I. Salman, B. Turhan, P. Rodríguez, and P. Ralph, "Cognitive biases in software engineering: A systematic mapping study," *IEEE Transactions on Software Engineering*, vol. 46, no. 12, pp. 1318–1339, 2018.

[22] T. A. S. Muneer, "Leveraging collective intelligence in agile sprint planning: A comparative study of LLM architectures," in *2025 International Conference on Smart & Sustainable Technology (INCSST).* IEEE, 2025, pp. 1–5.

[23] R. E. Nisbett and T. D. Wilson, "Telling more than we can know: Verbal reports on mental processes," *Psychological review*, vol. 84, no. 3, pp. 231–259, 1977.

[24] R. Parasuraman and D. H. Manzey, "Complacency and bias in human use of automation: An attentional integration," *Human factors*, vol. 52, no. 3, pp. 381–410, 2010.

Appendix A

Interview Guide

Part 1: Background (approximately 8 minutes)

1. Can you tell me about your current role and how long you have been working in Agile teams?
2. Walk me through how your team runs a typical sprint planning session: who is involved, how long it takes, and how you reach the final commitment.
3. What are the most common difficulties your team faces during sprint planning?
4. Does your team use any AI tools to support sprint planning decisions? If yes, what does that look like in practice?
   [If no AI experience: "Has AI come up in any conversation about your planning process, even if not adopted yet?"]

Part 2: Vignette Scenarios (approximately 12 minutes)

Do not tell participants what you are looking for. Read each scenario naturally and let them talk.

Vignette A: Anchoring Bias

Read aloud: "Your team is in sprint planning. An AI tool has analysed your backlog and past velocity data and recommends committing to 42 story points this sprint. Your team's average over the last five sprints has been 34 points. The AI tool explains that three similar stories were completed faster than expected recently, which is why it suggests a higher commitment."

5. What would you and your team do with this recommendation? Accept it, adjust it, or ignore it? Walk me through your thinking.
6. What information would you need before feeling confident making a decision here?
7. If your team's discussion started revolving around the number 42 rather than your usual 34, would that concern you? Why or why not?

8. Who would have the most influence over the final decision: the AI tool recommendation, the Scrum Master, the developers, or someone else?

Vignette B: Algorithm Aversion

Read aloud: "The same AI tool flags that a story your team has already agreed to include this sprint will likely take 60% longer than estimated, based on similar past stories. The developer assigned to that story disagrees. They say they know this part of the codebase well and are confident in the original estimate."

9. How would your team handle this? Side with the AI tool, the developer, or find a middle ground?

10. Have you ever been in a situation where a tool's recommendation conflicted with a team member's judgement? What happened?

11. What would make you more willing to trust the AI tool's warning in this case?

Part 3: Trust (approximately 10 minutes)

12. When you receive a recommendation from an AI tool during planning, what makes you trust it enough to act on it?

13. Can you describe a specific time when you trusted an AI tool recommendation and it turned out to be wrong? How did that change how you used the tool afterwards?
[If no AI experience: "What would make you trust or distrust an AI tool recommendation? What would it need to show you?"]

14. How important is it that an AI tool explains its reasoning, not just the output but why it made that recommendation?

15. Is there any planning decision you would never delegate to an AI tool, no matter how accurate it was? What and why?

16. What would an AI tool need to show you, in terms of transparency, track record, or explanation, before you felt comfortable letting it handle a planning decision without your direct review?

17. If you imagine sprint planning where AI tools and humans work together well, what does that actually look like? Who does what?

Closing

18. Is there anything about AI in sprint planning that you think researchers typically miss or get wrong?

Appendix B

Codebook

The codebook is organised into three groups corresponding to the three main analytical themes that emerged from the data. Group A covers anchoring codes, capturing the mechanism of number-first contamination, how it manifests in reasoning, and the safeguard practitioners developed in response. These codes address RQ1a and RQ1b and mapped to the task characteristics and teaming characteristics dimensions of HACO in Phase B. Group B covers algorithm aversion codes, capturing the full spectrum from unjustified rejection to calibrated scepticism, including the facilitation safeguards used to counter unjustified dismissal. These codes also address RQ1a and RQ1b and mapped to teaming characteristics in HACO. Group C covers trust calibration codes, capturing what drives trust up and down and how the facilitator-as-mediator model works in practice. These codes address RQ2 and mapped to the trust dimension of HACO. The three groups together produced the three model components: Group A and B codes produced the role zone definitions and interaction principles, Group C codes produced the trust-related principles and risk conditions.

Anchoring codes

A1: Number-first contamination. AI tool figure shared before independent team estimation shapes subsequent reasoning. Coded when a participant describes the risk of sharing a number early, or when their own vignette reasoning begins from the AI tool's figure.

A2: Reference-point adjustment. Participant reasons from the AI tool's figure rather than forming an independent judgement.

A3: Number withholding (safeguard). Deliberate practice of keeping the AI tool's figure private until after team estimation.

Algorithm aversion codes

B1: Developer authority assertion. Developer invokes domain expertise to dismiss an AI tool flag without articulating a specific basis.

B2: Examine both signals (safeguard). Facilitator requires both AI tool and developer to articulate their basis before the team decides.

B3: Pattern match scepticism. Participant questions whether the AI tool's historical patterns apply to the current context.

B4: Legitimate contextual dismissal. Override based on a concrete identifiable reason that the AI tool's evidence does not apply.

Trust calibration codes

C1: Transparency drives trust. Trust increases when the AI tool explains the basis for its recommendation.

C2: Track record drives trust. Trust updated by past experience of AI tool accuracy or inaccuracy.

C3: Human final authority. Final commitment decision must remain with the team regardless of AI tool capability.

C4: Facilitator as mediator. AI tool informs facilitator before session; facilitator decides what to bring and how to frame it.

C5: Contextual applicability assessment. Trust calibrated by assessing whether AI tool patterns apply to current specific conditions.

Appendix C

Validation Walkthrough Materials

Outreach message sent to P6, P7, and P8

The following message was sent to the three validation participants via WhatsApp. Each participant had already completed a full interview and was familiar with the research context.

> Hi [name], hope you're doing well. Following up from our interview last month. I've built the conceptual model from the research and would love your quick feedback on it. Here's a quick summary of the model:
>
> **Component 1 — Role Zones.** Zone 1: AI does the data work before the session — velocity trends, risk flagging, complexity flagging. Never speaks to the team directly. Zone 2: Facilitator filters AI output and decides what to bring into the room and how to frame it. Zone 3: Human only — final sprint commitment, override decisions, accountability stays with the team.
>
> **Component 2 — Interaction Principles.** (1) Number withholding — AI figure never shown to team before independent estimation. (2) Reasoning transparency — AI must explain which stories it is drawing on and why. (3) Examine both signals — when AI and developer disagree, both must articulate their basis. (4) Explicit override justification — overrides must state a concrete specific reason. (5) Human final authority — team always makes the final commitment.
>
> **Component 3 — Risk Checklist.** High risk: AI number shared before estimation, first sprint with new tool, high pressure cycle, senior judgement unchallenged, no explanation of basis. Moderate risk: unfamiliar AI data sources, codebase changes, long gap since similar work, developer confidence unarticulated.
>
> Three questions: (1) Looking at the three zones, does this reflect how you think AI should be involved in sprint planning? Anything missing or unrealistic? (2) Looking at the five principles, any you would not follow in practice, or any missing? (3) Looking at the risk checklist, does this match your experience? Anything to add or remove? Feel free to just reply here, no need for a call if that's easier.

P6 response

Q1 — Three zones: "The zone structure makes sense to me. Zone 1 is the right place for AI — it does its background work before the session and I decide what to bring in. Zone 2 is where I sit anyway — I am the filter between the tool and the team. Zone 3 is non-negotiable, the team calls the commitment, not the tool. One thing I would add is that Zone 1 should explicitly include flagging stories where the backlog is not groomed enough to estimate — that is a pre-session signal the AI could usefully surface."

Q2 — Five principles: "Number withholding is exactly right and I would go further — I would not even tell the team a recommendation exists until after they have estimated. Transparency is non-negotiable for me, a black box recommendation in planning creates false authority without a counter argument. Examining both signals is how I already run disagreements. Override justification is good — I tell my team that saying the tool does not know our context is not a reason, it is a starting point for a reason. Human final authority, obviously yes, always."

Q3 — Risk checklist: "The five high risk conditions are all real. The one I would add is overconfidence after a good sprint run — I had a PO who would open planning by announcing a target and the team would work backwards from it every time. That social pressure condition is missing. The moderate risks look right."

P7 response

Q1 — Three zones: "Yes this reflects how I think about it. The zone boundary between 1 and 2 is important — the AI should never be presenting its output directly to the team, it should always go through me first. I tried the other way with Momentum and it contaminated the room every time. The only thing I would refine is Zone 2 — I would make explicit that the facilitator decides not just what to bring but when during the session to bring it. Timing matters as much as content."

Q2 — Five principles: "Number withholding is the most important one and I am glad it is first. I learned this the hard way — the morning the recommendation landed in Slack before planning was a disaster. Transparency is table stakes, I will not bring something into the room I cannot explain. Examine both signals is what I do — my job is to make sure the team has looked at both properly, not to pick a side. Override justification is good but 2 out of 9 is low evidence — I would keep it in because it is the right behaviour even if participants did not articulate it clearly. Human final authority absolutely — the sprint commitment is a social contract, accountability requires authorship."

Q3 — Risk checklist: "All of these are real. The one I would add from my own experience is what I call sprint inertia — when previous good sprints create collective overconfidence and the team commits based on mood rather than data. That is its own risk condition separate from high pressure cycles. Also I would flag that the first sprint with a new AI tool is extremely high risk in both directions — people either over-trust it because it is new or dismiss it entirely. Both happened with Momentum."

P8 response

Q1 — Three zones: "This maps very closely to how I already work. The AI does its analysis, it comes to me, I decide what is worth raising and how to frame it, the team estimates and commits. Zone 2 is the right concept but I would describe it as the AI being background, not foreground — it is a well-prepared assistant who has done the research, not a co-facilitator whose recommendations need to be addressed in real time. The commitment stays entirely with the team."

Q2 — Five principles: "Number withholding is exactly right — if the tool's number is in the room before the team estimates, estimation becomes a response to the tool rather than a genuine judgement about the work. Transparency is important, the tools I use that show me which stories they are drawing on are the ones I trust because I can calibrate whether the historical data is still representative. Examine both signals is what I do in practice — I treat it as examine rather than picking a side. Override justification I would phrase as being explicit when I am overriding and why — I do not want the tool to become a shield where decisions feel justified by the AI said so. Human final authority — there is a real difference between a team saying we commit to this and a team saying the tool said we can do this so we did not disagree. The first means they will deliver it."

Q3 — Risk checklist: "These look right. I would add one condition to the moderate list — when the AI is drawing on data from before a significant team composition change. I had a case where the tool was flagging based on velocity data from when we had two more developers. The historical context had shifted but the tool did not know that. That needs to be in the checklist."

Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden